

Quant & AI Maths — Living Course Plan

Owner: Tatenda · **Started:** August 2026 **Goal:** Rebuild maths from A-level (last touched ~16 years ago) to a working level for quantitative trading and AI/ML. **Method:** Code-first — every concept gets implemented in Python/NumPy. Target ~45–60 min daily; consistency beats binges.

How to use this file with Claude Code: This is the master plan. Month 1 is fully detailed; Months 2+ are skeletons. At the start of each month, ask Claude Code to read this file plus the Progress Log and expand the next month to the same detail as Month 1. Tick checkboxes as you go. Keep code in `exercises/week-NN/`, bigger builds in `projects/`, reflections in `notes/`.

Suggested repo layout

```
maths-course/  
├─ PLAN.md           ← this file  
├─ exercises/  
│   └─ week-01/  
├─ projects/  
└─ notes/
```

Full roadmap (skeleton)

Phase	Months	Topic	Status
0	1	A-level refresh + tooling	 in progress
1	2–3	Linear algebra	☐
2	4	Multivariable calculus + optimization	☐
3	5–6	Probability	☐
4	7	Statistics	☐
5a	8+	Quant fork: time series, stochastic processes, Monte Carlo	☐

5b	8+	AI fork: matrix calculus, information theory, convex optimization	☐
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Phases 5a/5b can run in parallel or one after the other — decide when you get there.

Month 1 — A-level refresh + tooling (detailed)

Goal: shake off the rust on algebra, functions, trig, and calculus, and set up the coding workflow that carries the whole course.

Daily shape: ~30 min lessons + pen-and-paper problems, ~20 min implementing the day's idea in code.

Week 1 — Algebra, functions & setup

- Set up the repo, Python env, Jupyter, NumPy + Matplotlib
- Take a diagnostic: Math Academy placement, or Khan Academy “course challenge” for Algebra II — record the result in the Progress Log
- Refresh: solving equations and inequalities, exponentials & logarithms, function transformations
- **Code:** plot function families (polynomials, exp, log); write a `bisection_solve(f, a, b)` root-finder
- **Checkpoint:** solve 10 mixed algebra/log problems without notes

Week 2 — Trig, sequences & series

- Refresh: radians, sin/cos/tan, the key identities
- Refresh: arithmetic & geometric sequences and series, sum formulas
- **Code:** compound interest and discounted cash flow via geometric series (first finance link); Taylor-approximate $\sin(x)$ and plot it against the true curve
- **Checkpoint:** derive and code the geometric series sum; explain why it converges when $|r| < 1$

Week 3 — Differentiation

- Refresh: the limit idea, derivative rules, chain/product/quotient rules
- Refresh: stationary points, max/min word problems

- **Code:** finite-difference numerical derivative; 1-D gradient descent to find a function's minimum (first ML link)
- **Checkpoint:** hand-differentiate 8 functions and verify each one numerically in code

Week 4 — Integration + first probability

- Refresh: integration as antiderivative and as area; definite integrals
- Intro: sample spaces, expected value (A-level stats territory)
- **Code (mini-project):** Monte Carlo integration — estimate π by random sampling, then integrate an arbitrary $f(x)$; plot error vs. sample count (first quant link)
- **Checkpoint / Month-1 milestone:** write `notes/month-01.md` — what came back easily, what didn't, and how the daily rhythm felt

Month 1 resources

- **Khan Academy** (free): Algebra II, Trigonometry, AP Calculus AB units 1–6
 - **Math Academy** (paid, adaptive): Mathematical Foundations sequence — strong option if you want the placement test to drive everything
 - **3Blue1Brown** — *Essence of Calculus*, watch alongside Weeks 3–4
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Months 2+ (skeletons — expand with Claude Code)

Months 2–3 — Linear algebra

Vectors, matrices, linear maps, eigenvalues/eigenvectors, SVD. Anchors: 3Blue1Brown *Essence of Linear Algebra*, MIT 18.06 (Strang), *Mathematics for Machine Learning* ch. 2–4. Projects: matrix ops from scratch, then PCA on real stock or crypto returns.

Month 4 — Multivariable calculus & optimization

Partial derivatives, gradients, Jacobians, chain rule, Lagrange multipliers, gradient descent variants. Projects: 2-D gradient descent visualizer; a tiny autodiff engine.

Months 5–6 — Probability

Distributions, conditional probability, Bayes, law of large numbers, CLT. Anchor: Blitzstein's Stat 110 (free lectures + book). Projects: simulate the CLT; a Bayesian A/B test.

Month 7 — Statistics

Regression from scratch (OLS via your linear algebra), hypothesis testing, maximum likelihood. Project: OLS with diagnostics on market data.

Months 8+ — The fork

- **Quant:** time series (stationarity, ARIMA, GARCH, cointegration), random walks & Brownian motion, a taste of Itô, Monte Carlo pricing. **Capstone:** pairs-trading backtest — crypto data is an easy, familiar sandbox.
 - **AI:** matrix calculus, information theory (entropy, KL divergence, cross-entropy), convex optimization. **Capstone:** a NumPy-only neural net trained end-to-end on a small dataset.
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Reference shelf

- *Mathematics for Machine Learning* — Deisenroth, Faisal, Ong (free PDF)
- *Introduction to Probability* — Blitzstein & Hwang
- *Introduction to Linear Algebra* — Strang
- Quant side, for later: *Paul Wilmott Introduces Quantitative Finance*; Zhou's *A Practical Guide to Quantitative Finance Interviews* (the "green book")

Progress log

Date	What I did	Time	Notes / blockers

Prompts for future Claude Code sessions

- "Read PLAN.md and the Progress Log; expand Month 2 into a week-by-week plan at the same detail as Month 1."
- "Generate this week's exercise set into `exercises/week-NN/`, with solutions in a separate file I can't see by accident."
- "Quiz me on last week's checkpoints before unlocking this week."
- "Review my `notes/month-NN.md` and adjust next month's pacing accordingly."

